

Apurva Aggarwal

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EDUCATION

Michigan State University

Bachelor of Science in Computer Science, Minor in Business

East Lansing, MI

May 2027

- **GPA:** 3.95 — **Dean's List (Top 1% amongst 500)**
- **Relevant Coursework:** Cloud Development, Biometrics, Artificial Intelligence, Big Data Analysis

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, JavaScript, R, Assembly **Data & ML:** Pandas, NumPy, scikit-learn, TensorFlow, PyTorch
Systems & Cloud: Azure, Docker, Git, REST APIs, Microservices **Analytics & Visualization:** Tableau, Power BI

EXPERIENCE

Data Analytics Engineering Intern – Tesla, Inc., Residential Energy | Palo Alto, CA Jan 2026 – Aug 2026

- Engineered a production-scale **entity resolution pipeline** using TF-IDF weighting, Levenshtein similarity, and geo-aware reranking, achieving **90%+ accuracy** while reducing manual review by **280 hours per 10K records**
- Built internal **AI workflow systems** across **16 sales workflows**, translating natural-language requests into automated sales enablement processes through prompt orchestration and enterprise AI integrations
- Owned executive-facing **business reporting** for Tesla's residential solar products, defining KPI frameworks and tracking product performance and rebate program adoption across weekly, monthly, and quarterly reviews
- Built **decision-support analytics** for 20+ Sales and Sales Ops stakeholders, transforming CRM and operational data into performance benchmarks and strategic insights used for account prioritization and sales performance management

Technology Consulting Intern – Ernst & Young (EY) | Singapore May 2025 – Aug 2025

- Led client-facing **UAT** for migration of 10K+ records to a cloud-based platform, identifying critical defects prior to release
- Automated migration validation workflows using **SQL (SSMS)**, reducing manual QA effort by 50%
- Migrated legacy file storage to **Azure Blob Storage**, enabling secure infrastructure for a \$13M enterprise system
- Managed sprint workflows and defect tracking in **Azure DevOps** to ensure on-time module delivery

Data Analyst – The Global Career Accelerator | Remote Aug 2024 – Dec 2024

- Analyzed **30M+ engagement records**, uncovering 5× higher engagement rates for photo posts
- Designed statistically powered **A/B experiments**, performing hypothesis testing, power analysis, and MDE calculations to evaluate campaign and conversion rate changes
- Built **KPI dashboards** in SQL and Tableau, standardizing reporting pipelines for executive stakeholders
- Led **cohort-based segmentation** across multi-industry datasets to quantify growth and churn drivers

PROJECTS

Argus | Python, Streamlit, PostgreSQL, SQLAlchemy, GitHub Actions Jun 2026

- Built an **AI infrastructure intelligence platform** tracking 50+ companies across AI, semiconductor, and quantum
- Developed automated **ETL pipelines** using GitHub Actions for market data, SEC filings, earnings events, and news
- Designed custom **scoring models** and thematic indices to identify pullback opportunities and sector-wide trends
- Architected a modular analytics framework supporting watchlists, alerts, provider abstractions, and cloud deployment

FarmX (MHacks 2024 Winner) | Python, scikit-learn, Streamlit Sept 2024

- Built an end-to-end ML pipeline using **Random Forest regression** to optimize NPK nutrient levels, reducing waste by 25%
- Engineered data preprocessing and feature engineering workflows using **Pandas and NumPy**
- Developed a lightweight backend in Streamlit with caching optimization, improving response time by **30%**
- Integrated **weather and soil APIs** to enable real-time adaptive recommendations

Celebrity Face Recognition | PyTorch, CNNs, Transfer Learning Dec 2025

- Implemented a transfer learning pipeline using a pretrained **AlexNet** model to classify a 17-class celebrity dataset
- Achieved **57.35% accuracy** using k-fold cross-validation on a high intra-class similarity dataset
- Applied data augmentation and dropout regularization to improve model generalization and reduce overfitting
- Analyzed misclassifications through confusion matrices and error inspection to guide iterative model refinement

Micro Foods Market (Backend) | Docker, Flask, REST APIs, SQLite, JWT Apr 2025

- Designed a **microservices backend** with 5 Flask services for auth, orders, search, logs, and product management
- Implemented REST APIs and **JWT-based authentication** for secure distributed transaction workflows
- Configured Docker networking and service orchestration for **cloud deployment readiness**

LEADERSHIP & INVOLVEMENT

SpartaHack – Co-Director for 600+ attendee student-run hackathon

Resident Assistant – Mentored and supported 80+ students in a residential community

International Students Association – Director of Technical Affairs, building centralized affiliate web platform

Instagram Content Creator – Produced music content reaching 20K+ views